

SOLUTIONS

1. (1 point) A differentiable function f is defined on the closed interval $[-2, 4]$. Which of the following statements must be true?

- A. If f has a local maximum at $x = 4$, then $f'(4) = 0$
- B. If $f'(c) = 0$ for some $c \in (-2, 4)$, then f has a local extremum at $x = c$
- C. If f has a local minimum at $x = 2$, then $f''(2) > 0$
- D. If f has a local maximum at $x = 1$, then $f'(1) = 0$

Solution: By Fermat's Theorem, if f has a local extremum at an interior point c , then $f'(c) = 0$. Since $x = 1$ is in the interior $(-2, 4)$, (d) must be true.

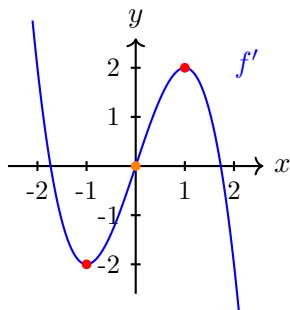
(a) is false: Fermat's Theorem does not apply at endpoints, so a differentiable function can have a local maximum at $x = 4$ without $f'(4) = 0$.

(b) is false: $f'(c) = 0$ does not guarantee a local extremum. For example, $f(x) = x^3$ has $f'(0) = 0$, but $x = 0$ is neither a local maximum nor minimum.

(c) is false: the Second Derivative Test requires $f'(c) = 0$ as a hypothesis and does not apply when $f''(c) = 0$. For instance, $f(x) = x^4$ has a local minimum at $x = 0$ with $f''(0) = 0$.

(d) is correct.

2. (1 point) The graph of f' , the derivative of a twice-differentiable function f , is shown below. How many inflection points does the graph of f have on the interval shown?



- A. 0
- B. 1
- C. 2
- D. 3

Solution: An inflection point of f occurs where f'' changes sign, i.e., where the slope of f' changes sign. This corresponds to local extrema of f' (not zeros of f').

From the graph, f' has a local minimum at $x = -1$ and a local maximum at $x = 1$. At each of these, the slope of f' changes sign, so f'' changes sign. Therefore f has 2 inflection points.

The zero of f' at $x = 0$ is where f has a local extremum, not an inflection point.

(c) is correct.

3. (1 point) Let $f(x) = x^3 - 6x^2 + 9x$. On the interval $[0, 4]$, the Mean Value Theorem guarantees the existence of a point c in $(0, 4)$ for which $f'(c)$ equals the average rate of change of f on $[0, 4]$. What is the value of c ?

A. 1

C. $2 + \frac{2\sqrt{3}}{3}$

B. 2

D. $2 - \frac{2\sqrt{3}}{3}$

Solution: Average rate of change:

$$\frac{f(4) - f(0)}{4 - 0} = \frac{(64 - 96 + 36) - (0)}{4} = \frac{4}{4} = 1.$$

Solve $f'(c) = 1$:

$$f'(x) = 3x^2 - 12x + 9 = 1 \implies 3x^2 - 12x + 8 = 0 \implies x = \frac{12 \pm \sqrt{144 - 96}}{6} = \frac{12 \pm 4\sqrt{3}}{6} = 2 \pm \frac{2\sqrt{3}}{3}.$$

Only $c = 2 + \frac{2\sqrt{3}}{3} \approx 3.15$ lies in $(0, 4)$.

(c) is correct.

4. (1 point) The derivative of a function h is given by $h'(x) = (x + 2)(x - 1)^2(x - 4)$. At which of the following values of x does h have a local minimum?

A. $x = -2$

C. $x = 4$

B. $x = 1$

D. h has no local minimum

Solution: Sign chart for h' using the factors $(x + 2)$, $(x - 1)^2$, and $(x - 4)$:

	$(-\infty, -2)$	$(-2, 1)$	$(1, 4)$	$(4, \infty)$
$(x + 2)$	−	+	+	+
$(x - 1)^2$	+	+	+	+
$(x - 4)$	−	−	−	+
$h'(x)$	+	−	−	+

At $x = 4$, h' changes from negative to positive, so h has a local minimum. At $x = -2$, h' changes from positive to negative, giving a local maximum. At $x = 1$, h' does not change sign.

(c) is correct.

5. (1 point) Let $f(x) = x^4 - 4x^3$. Use the First Derivative Test to classify the critical points of f .

Solution: $f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$.

Critical points: $x = 0$ and $x = 3$.

Sign chart for f' :

	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
$4x^2$	+	+	+
$(x - 3)$	-	-	+
$f'(x)$	-	-	+

At $x = 3$: f' changes from negative to positive, so f has a local minimum at $x = 3$.

At $x = 0$: f' does not change sign, so $x = 0$ is neither a local maximum nor a local minimum.

6. (2 points) Find the absolute maximum and absolute minimum values of $g(x) = x + 2\sin x$ on the closed interval $[0, \pi]$. Show your work.

Solution: Differentiate: $g'(x) = 1 + 2\cos x$.

Set $g'(x) = 0$: $\cos x = -\frac{1}{2} \implies x = \frac{2\pi}{3}$ (the only solution in $(0, \pi)$).

Evaluate at all candidates:

x	$g(x)$
0	0
$\frac{2\pi}{3}$	$\frac{2\pi}{3} + 2\sin\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sqrt{3}$
π	π

Comparing: $0 < \pi < \frac{2\pi}{3} + \sqrt{3}$, since $\sqrt{3} > \pi - \frac{2\pi}{3} = \frac{\pi}{3}$.

Absolute maximum: $\frac{2\pi}{3} + \sqrt{3}$ at $x = \frac{2\pi}{3}$. Absolute minimum: 0 at $x = 0$.